

Requirement Analysis

Date	06 NOVEMBER 2025
Team ID	NM2025TMID03796
Project Name	LEASE MANAGEMENT

1. Introduction

The Requirement Analysis Phase identifies and defines all system functionalities, performance needs, and technical requirements for the Lease Management System.

It helps to ensure that the project meets user expectations by gathering and documenting what the property owners, tenants, and administrators need from the system.

This phase also clarifies functional, non-functional, and system-level requirements that will guide the design and development process.

2. Purpose

The purpose of requirement analysis is to:

- Understand the business needs of users.
- Translate user expectations into clear system requirements.
- Define the data flow, functional modules, and technology stack.
- Provide a solid foundation for system design and development.

3. Stakeholder Identification

Stakeholder	Role	Responsibility
Property Owner	System User	Creates and manages lease agreements, reviews payments.

Stakeholder	Role	Responsibility
Tenant	System User	Views lease details, uploads documents, and receives reminders.
Administrator	System Manager	Oversees all leases, generates reports, and maintains database.
Developer Team	Technical Team	Builds, tests, and deploys the system.
Mentor / Guide	Project Advisor	Validates project progress and outcomes.

4. Functional Requirements

Requirement ID	Functional Requirement Description
FR-01	The system shall allow property owners to create, view, edit, and delete lease agreements.
FR-02	The system shall enable tenants to upload supporting documents securely.
FR-03	The system shall send automated rent payment reminders to tenants.
FR-04	The system shall allow users to view active and expired leases.
FR-05	The system shall generate analytical reports and dashboards.
FR-06	The system shall support user authentication and access control.
FR-07	The system shall store all data in a centralized database.

5. Non-Functional Requirements

Requirement ID	Non-Functional Requirement Description
NFR-01	The system should be available 24/7 with 99% uptime.
NFR-02	The application should load within 3 seconds under normal conditions.
NFR-03	The system should be scalable to handle multiple users and properties.
NFR-04	All data must be stored securely with encryption.
NFR-05	The interface must be user-friendly and mobile-responsive.
NFR-06	The system should ensure role-based access for security.

6. Data Flow Diagram (DFD)

Level 0 – Context Diagram

- Entities: Tenant, Owner, Administrator
- System: Lease Management System
- Data Flow: Tenant provides lease information → System stores and tracks → Owner/Admin accesses reports and reminders.

Level 1 – Detailed Data Flow

1. Tenant uploads lease document → stored in database.

2. System validates and saves lease data.
3. System triggers rent reminder notifications.
4. Owner/Admin reviews payments and generates reports.

(You can include this as a diagram in your PDF using circles and arrows to represent flow.)

7. User Stories

User Type	User Story Description
As a tenant	I want to receive rent reminders so I can make timely payments.
As a property owner	I want to track lease renewals so I can manage properties efficiently.
As an administrator	I want to generate reports on active and expired leases for record keeping.
As a developer	I want to ensure data security to protect sensitive user information.

8. Solution Requirements

Category	Requirement
Software Requirements	Salesforce Platform / VS Code / JMeter / Postman
Hardware Requirements	Laptop or PC with minimum 8 GB RAM and stable internet
Database Requirements	Salesforce Object Database / SQL
User Interface Requirements	Dashboard, Login Page, Lease Entry Form

Category	Requirement
Performance Requirements	Must handle at least 200 concurrent users without delay

9. Technology Stack

Layer	Technology Used	Purpose
Frontend	HTML, CSS, JavaScript / Salesforce Lightning Components	User Interface
Backend	Apex (Salesforce) / Node.js	Application Logic
Database	Salesforce Objects / SQL	Data Storage
Testing Tools	JMeter, Postman	Performance and API Testing
Version Control	GitHub	Source Code Management
Deployment	Salesforce Cloud / Web Server	Hosting and Maintenance

10. Conclusion

The Requirement Analysis Phase establishes a clear understanding of what the Lease Management System must deliver.

By identifying user needs, defining functional and non-functional requirements, and mapping data flows, this phase ensures that the development team builds a robust, user-centered, and scalable system.

Proper requirement documentation minimizes risks, prevents misunderstandings, and sets the foundation for a successful project implementation.